PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2011

Candidate Name	

Civics Group						**INDEX NUMBER**					

H2 BIOLOGY **9648 / 02**

PAPER 2 CORE PAPER **13 Sept 2011**

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.
This document consists of 2 sections:

Section A
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B
Answer only **ONE** question.
Write your answer on the foolscap paper provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	**/ 10**
2	**/ 12**
3	**/ 10**
4	**/ 12**
5	**/ 10**
6	**/ 13**
7	**/ 13**
Section B	
8 OR 9	**/ 20**
TOTAL	**/ 100**

This document consists of **23** printed pages (inclusive of this page)

[Turn Over

Question 1

In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in table 1.1.

Table 1.1

Incubation period /min	**Colour intensity of the reaction mixture / arbitrary unit**		
	Tube A	**Tube B**	**Tube C**
	Protease + heated protein substrate	**Protease + unheated protein substrate**	**Protease + unheated protein substrate + raw bean extract**
0	10	10	10
30	4	7	9

Fig. 1.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

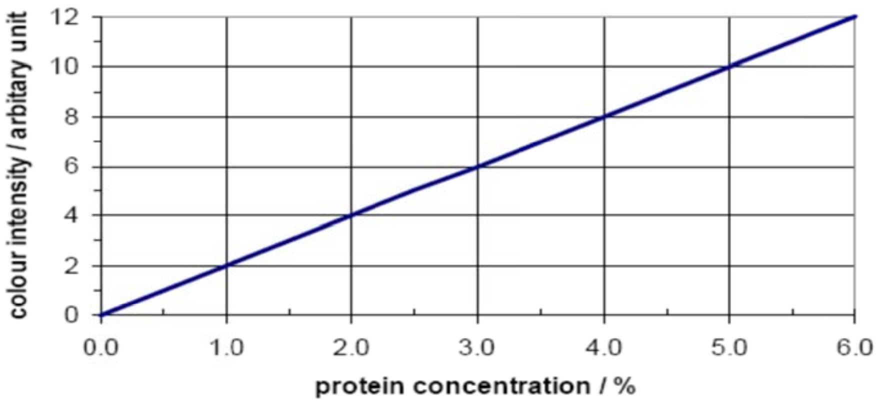

Fig 1.1

(a) With reference to Table 1.1 and Fig 1.1,

(i) Determine the protein concentration of the three tubes at time 0 min. [1]

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(ii) Explain which protein substrate is more digestible by protease. [2]

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(b) Based on your knowledge of protein structure, explain the effect of heat treatment in the digestibility of this protein substrate. [3]

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(c) Suggest the effect of raw bean extract on protease activity. [1]

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(d) Fig 1.2 shows the structure of a biochemical molecule, **W**.

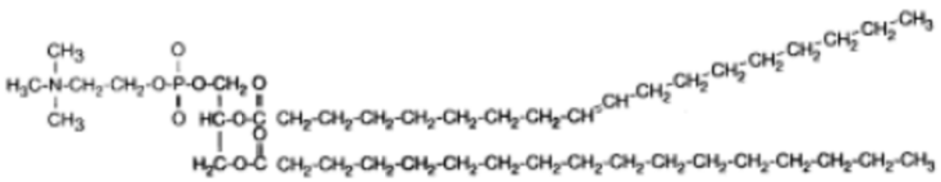

Fig 1.2

(i) Identify the molecule **W**. [1]

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(ii) Membrane lipids in reindeer hooves need to remain fluid at low temperatures. Based on the structure of molecule **W**, predict the composition of such membrane lipids. [1]

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(iii) The feather of certain aquatic birds such as ducks are covered with a protective layer of wax. Based on this information, predict why oil spills are so harmful to these birds. [1]

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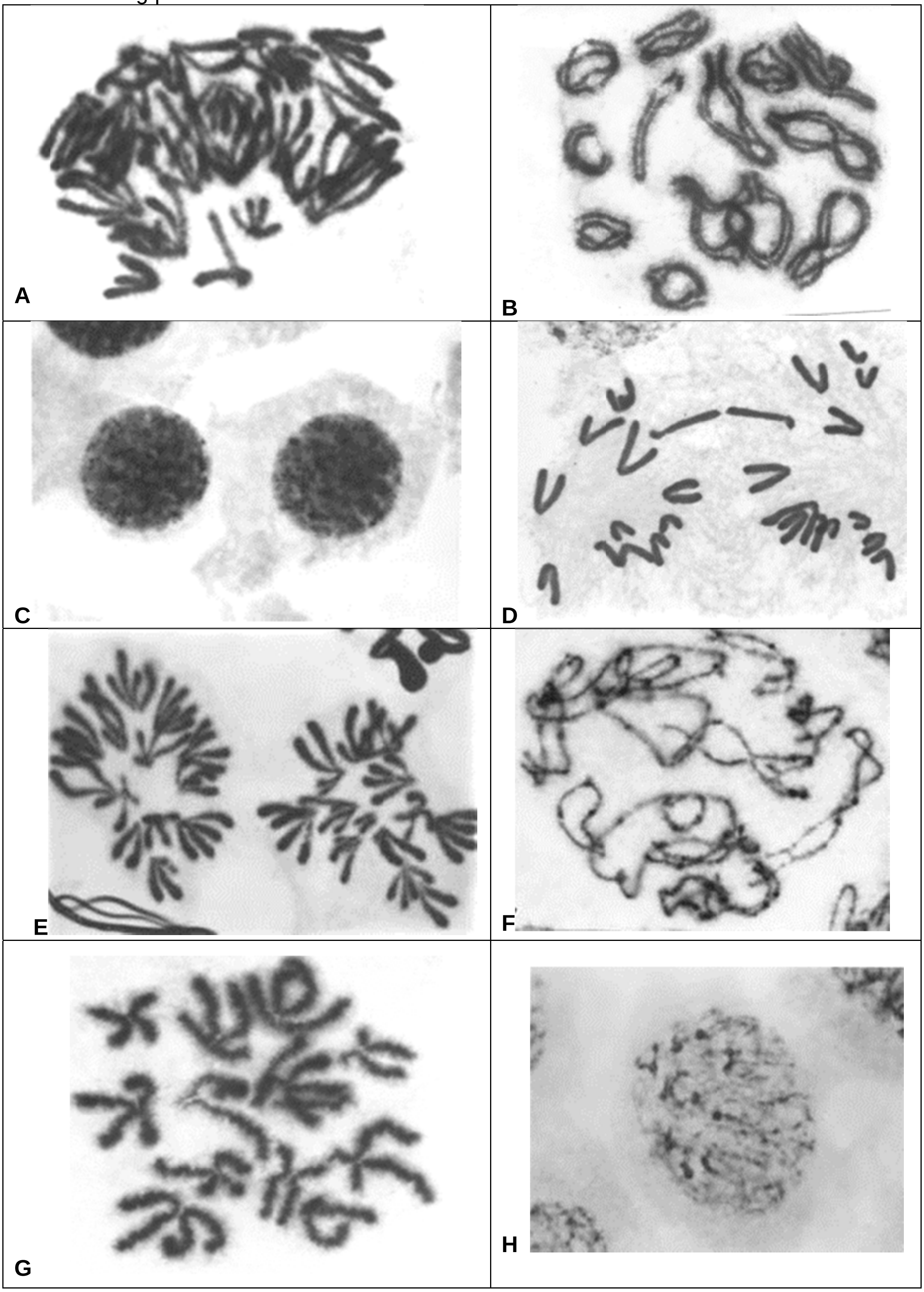
[Total: 10 marks]

Question 2

The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant.



(a) Micrographs **C** and **H** show the 5th and 6th stages in the sequence. In the boxes below, arrange the other letters shown on the micrographs to indicate the correct chronological sequence for this type of nuclear division. [2]

				C	**H**		

(b) What is the diploid number of this plant? With reference to at least 1 stage (name the stage) from the micrographs, explain how you arrive at your answer. [2]

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(c) Discuss the significance of stage **B**. [2]

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(d) Figure 2.3 shows changes in DNA content at different stages of the animal life cycle.

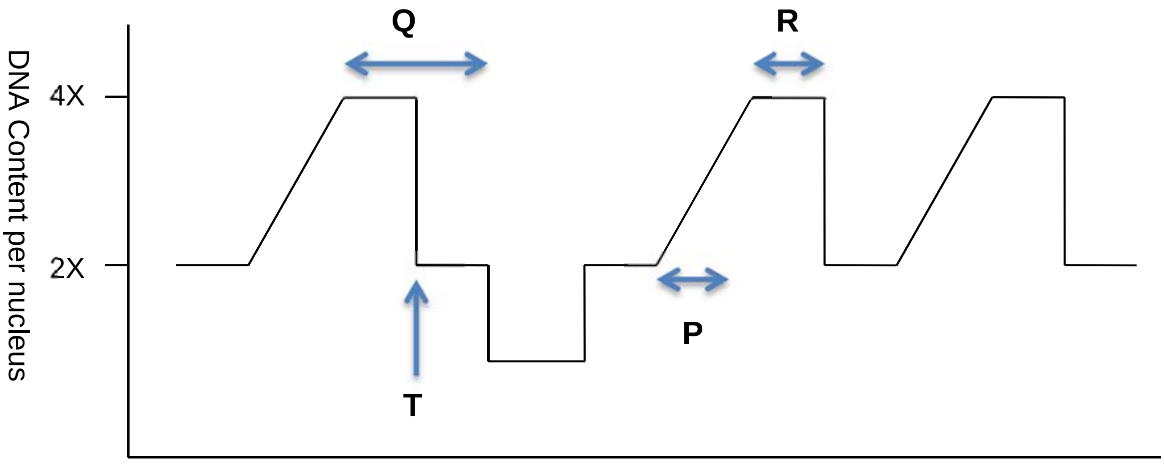

Figure 2.3

With reference to Figure 2.3,

(i) Identify the type of divisions and justify your answer. [2]

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Q ……………………………………………………………………………………

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(ii) Identify stage T. [1]

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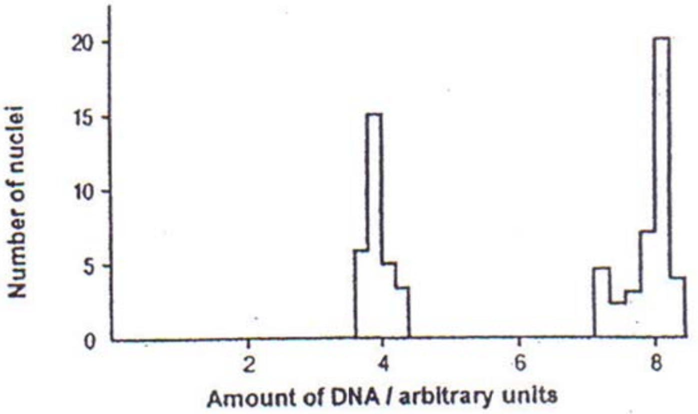
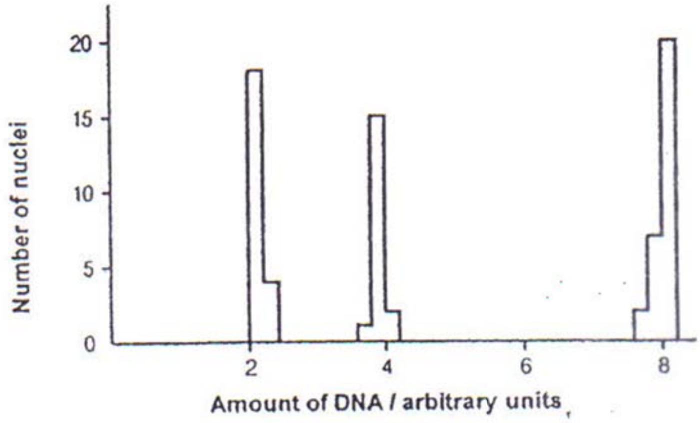
The formation of sperm cells involves mitosis and meiosis. The graphs below show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis.



Graph 1



Graph 2

With reference to the graphs,

(e)(i) Explain the evidence which shows that Graph 1 represents cells undergoing mitosis. [1]

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(ii) Explain how the events of meiosis result in three groups of nuclei in Graph 2. [2]

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Question 3
(a) Explain how the promoter of a eukaryotic gene influences transcription. [2]

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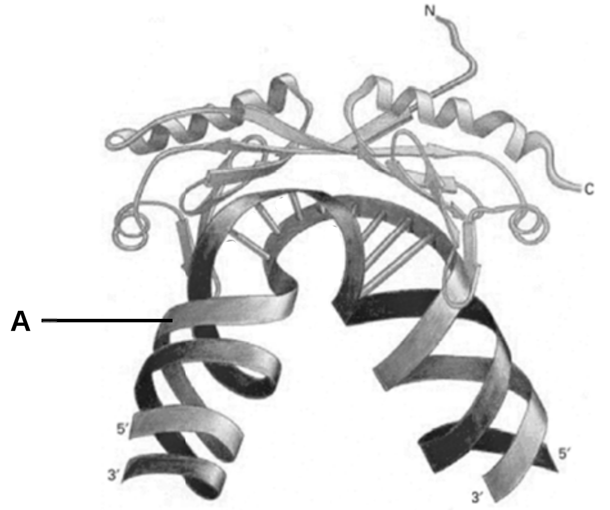
(b) Fig. 3.1 shows the binding of a general transcription factor to part of the promoter. This will help to recruit RNA polymerase to the promoter.



. 3.1

(i) With reference to Fig. 3.1, give **one** reason why the structure labelled **A** is part of the promoter and not the transcription factor. [1]

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(ii) Suggest why a substitution mutation at the promoter may result in the silencing of the gene controlled by the promoter. [1]

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(c) Describe how distal control elements can help increase the rate of transcription of a gene. [2]

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(d) Even though the control of gene expression occurs mostly at the level of transcription, control of gene expression also occurs at the level of post-translational modification.

Name an example of post-translational modification of a **specific** protein. [1]

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Human DNA has been analyzed and details of certain genes are shown in the table below.

Gene	**Gene size / kb***	**mRNA size / kb**	**Number of introns**
Insulin	1.7	0.4	2
Collagen	38.0	5.0	50
Albumin	25.0	2.1	14
Phenylalanine hydroxylase	90.0	2.4	12
Dystrophin	2000.0	17.0	50

*Kilobase pairs

(Source: William S Klug and Michael R Cummings, (2002), Concepts of Genetics, 7th Edition, Prentice Hall, page 314)

(e) (i) Calculate the average size of the introns for the albumin gene. (Show your working.) [1]

(ii) Briefly discuss the relationship between gene size and the number of introns. [1]

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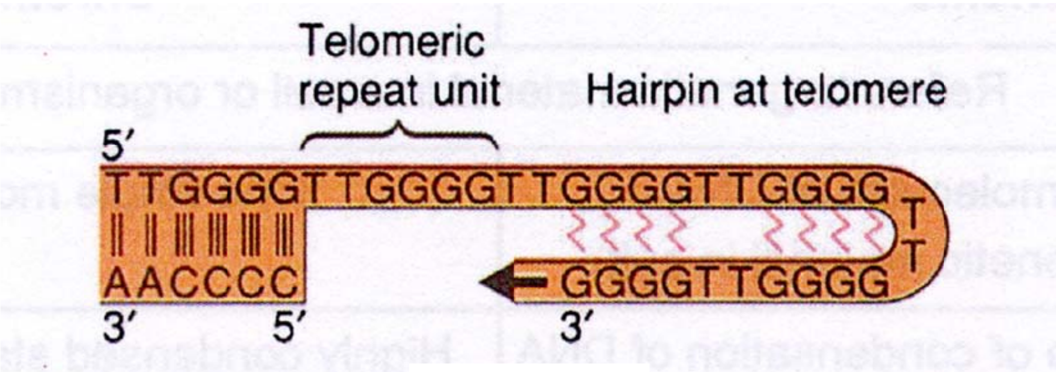
(iii) Determine the maximum number of amino acids that could be produced by translating the phenylalanine hydroxylase mRNA. (Show your working.) [1]

[Total: 10 marks]

Question 4

Figure 4.1 shows a diagram of a telomere. Telomeres posses a unique single stranded G-rich 3’ overhang that loops back to form a hairpin structure as shown.



g 4.1

(a) Describe the role of telomeres [1]

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(b) The telomerase enzyme is responsible for extending the telomere. It is essential in single celled organisms, while in humans and other vertebrates, it is active in reproductive cells.

(i) Explain how the 3’ overhang is formed at the end of a linear chromosome [1]

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(ii) Explain why regeneration of telomeres is not normally required in human body cells. [1]

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(d) The figure 4.2 below shows the sequence of events in a Ras cell cycle-stimulating pathway.

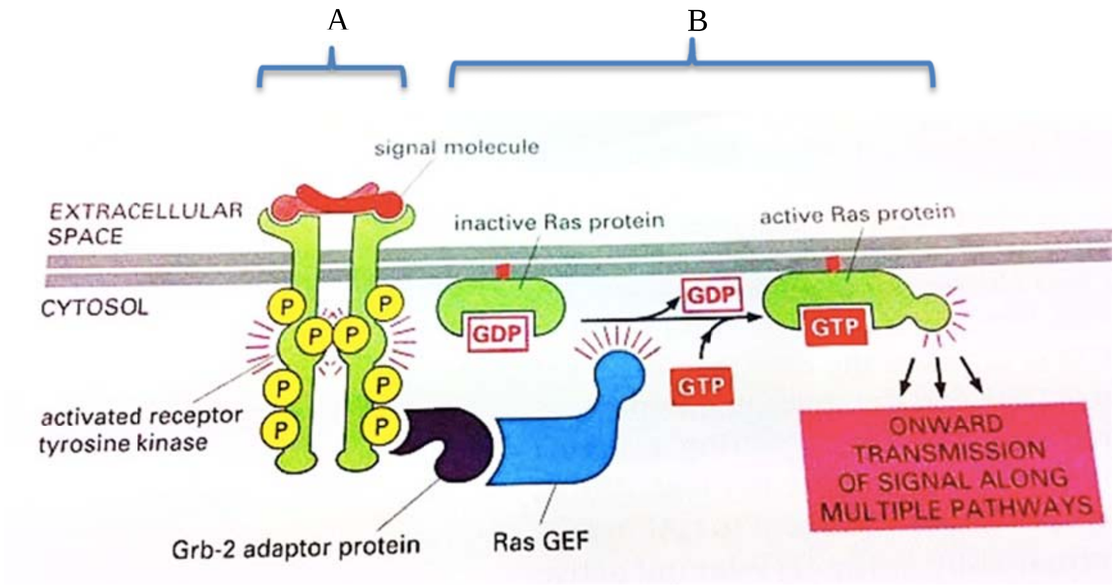

Fig. 4.2

(i) Briefly describe the stages A and B shown in Fig. 4.2. [3]

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(ii) Identify the signal molecule and explain why it cannot act directly on the DNA in the nucleus. [2]

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(iii) Explain how a small number of signal molecules can bring about a large cellular response. [2]

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(iv) With reference to Fig. 4.2, suggest how mutation in the Ras gene may bring about cancer. [2]

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[Total: 12 marks]

Question 5

In an investigation on the tomato genome, pollen grains from the anthers of a pure-bred plant with green leaves and hairy fruit were transferred by a scientist to the stigmas of a pure-bred plant with mottled (green and yellow) leaves and smooth-surfaced fruit. All the F_1 generation had green leaves and smooth fruit.

(a) Define what is meant by a pure-bred plant. [1]

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A test cross was then made between the F_1 plant and a pure-bred plant with mottled leaves and hairy fruit. The phenotypes of 100 offspring of the test cross were recorded and shown in **Table 5.1**.

Green leaves, smooth fruit	**Green leaves, hairy fruit**	**Mottled leaves, smooth fruit**	**Mottled leaves, hairy fruit**
8	46	40	6

Table 5.1

(b) Using suitable symbols, draw a genetic diagram to explain the results of the test cross. [5]

(c) Explain the deviation of the observed test cross results from the expected dihybrid test cross ratio. [2]

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A second scientist repeated the test cross between the same F_1 plant and pure-bred plant with mottled leaves and hairy fruit. However for this time, he only recorded the number of the 2 more common phenotypes of 100 offspring of the test cross as shown in **Table 5.2**.

Green leaves, hairy fruit	Mottled leaves, smooth fruit
52	44

Table 5.2

The formula for the chi-squared (χ^2) test is given as follows:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Degrees of freedom	**Probability**		
	0.10	**0.05**	**0.01**
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

(d)(i) This second scientist claimed that the observed test cross ratio of offspring can be simplified to 1:1. Perform the chi-squared (χ^2) test to verify his claim. [2]

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[Total: 10 marks]

Question 6

Dopamine is a chemical naturally produced in the body. In the brain, dopamine functions as a neurotransmitter, activating dopamine receptors on the post-synaptic membrane. Dopamine concentration affects brain processes that control body movement, emotional response and the ability to experience pleasure and pain.

(a) Describe the events that take place after the arrival of the pre-synaptic action potential that results in the activation of dopamine receptors on the post-synaptic membrane. [3]

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(b) State **two** reasons why nerve impulses can only travel in one direction across a synapse. [2]

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Fig. 6.1 shows the action potentials of a normal and mutant neurone. In the mutant neurone, a mutation in the gene coding for a channel protein results in a change to the action potential.

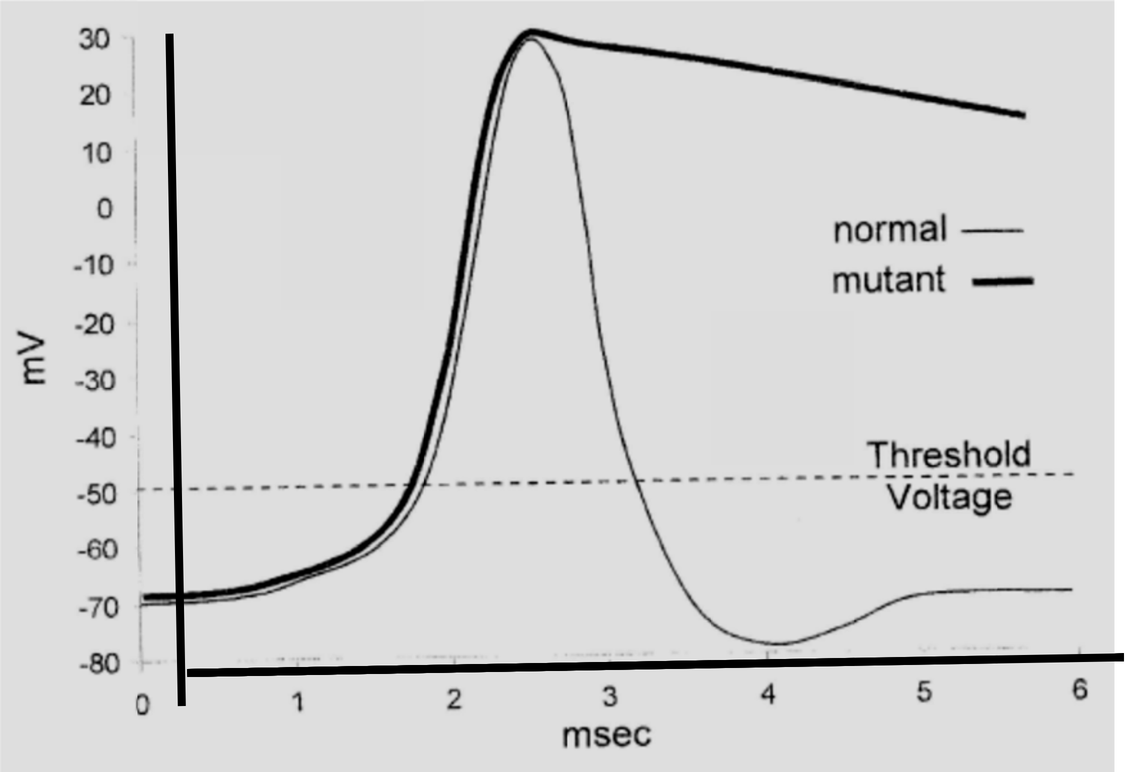

Fig. 6.1

(c) Describe how the resting potential of the normal neurone is maintained at -70mV. [2]

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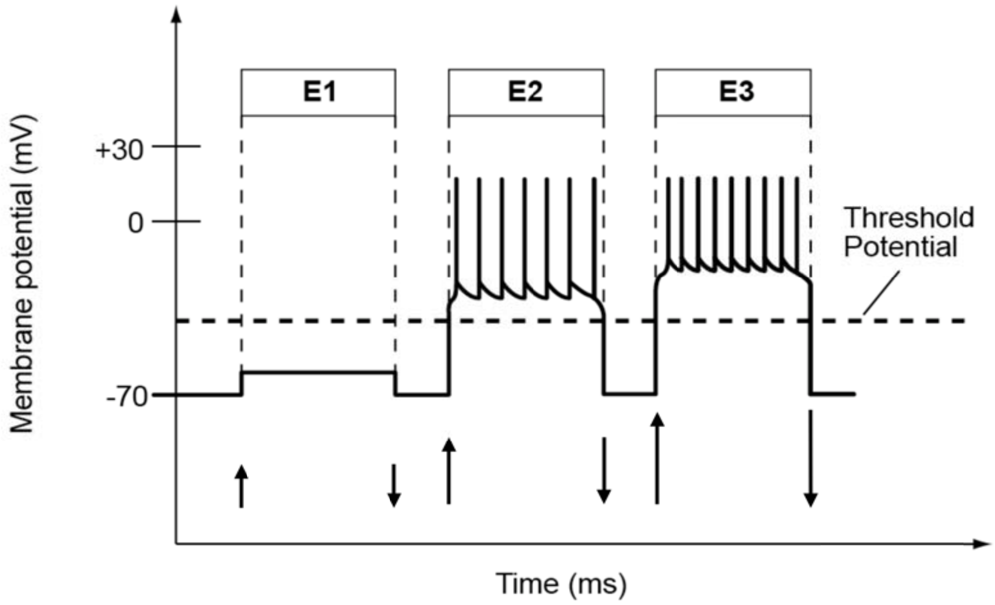
In a separate experiment, a group of researchers investigated how stimulus intensity affects nerve impulse transmission along the axon and synaptic transmission at the neuromuscular junction. They varied the strength of the stimulus and measured the extent of muscular contraction by recording the number of twitches produced per second. Electrodes were placed at different regions of the neurone to measure the changes in membrane potential generated after stimulation.

The experiment was repeated three times, **E1** to **E3**, using a different stimulus intensity each time. The results of the experiments are shown in **Fig. 6.4** and **Table 6.1**.

Key:
Upward arrow (↑): indicates point of stimulus application.
Downward arrow (↓): indicates point of stimulus cessation.
Length of arrow indicates strength of stimulus.



g. 6.2

Experiment	Number of twitches produced per second
E1	0
E2	7
E3	11

Table 6.1

(d) Describe the trend observed in **Fig. 6.2** for **E2** and **E3**. [2]

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(e)(i) State what you understand by the term ‘threshold potential’. [1]

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(ii) With reference to the results shown in **Fig. 6.2** and **Table 6.1**, explain how stimulus intensity affects the extent of muscular contraction. [3]

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[Total: 13 marks]

Question 7

Figure 7.1 below shows the classification of some primates. This classification is hierarchical and phylogenetic.

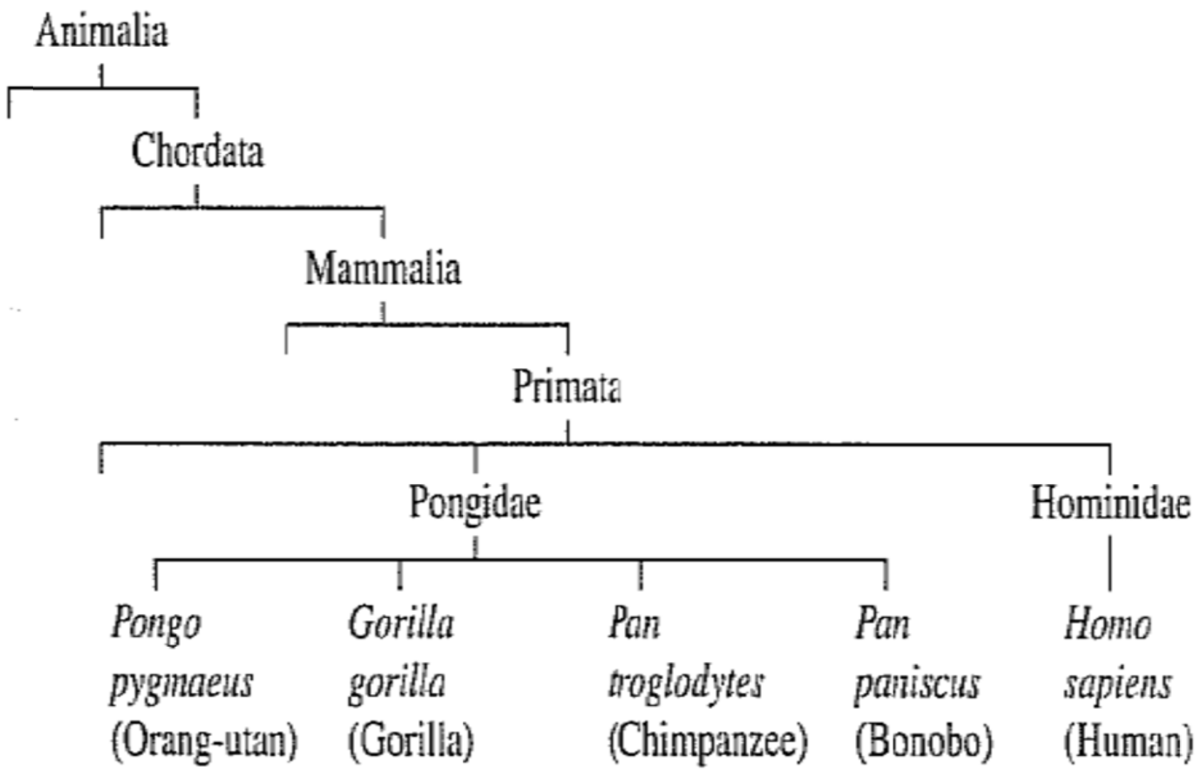

Figure 7.1. Classification of Some Primates

(a) Use the information from Figure 7.1,

(i) State which genus does the orang-utan belong? [1]

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(ii) State which order does the chimpanzee belong? [1]

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(b) Explain why this classification is both hierarchical as well as phylogenetic. [2]

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The Galapagos are a very isolated group of islands. Species of birds called finches live on these islands. Scientists think that the different species have evolved from a population of one species which managed to reach the islands thousands of years ago from South American mainland. Scientists used base sequences in the DNA of the finches to work out the evolutionary relationship between the species. Figure 7.2 shows the evolutionary relationship.

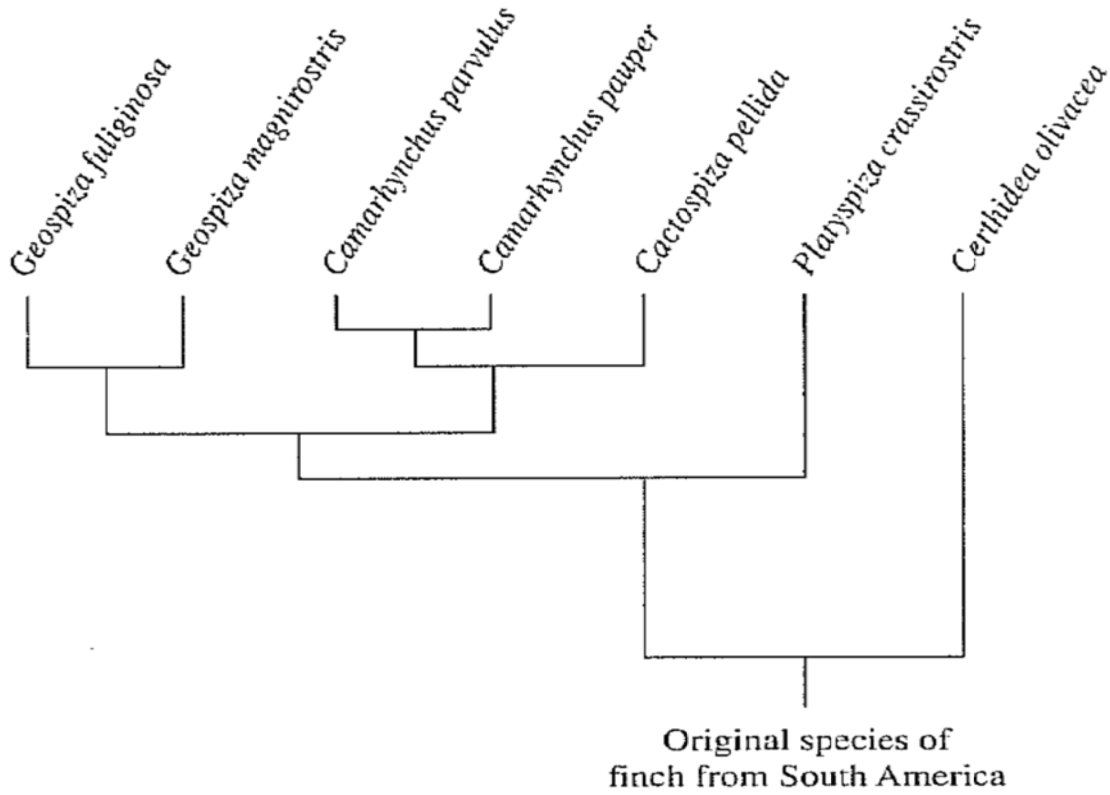

Fig. 7.2. Evolutionary Relationship between the Finches

(c) Using the information in the diagram, state the present-day species of finch that evolved first on the Galapagos islands? Explain your answer. [2]

Finch species:

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Explanation:

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(d) Explain why evaluating the DNA base sequences can work out the evolutionary relationship between the species of finches. [2]

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(e) The beak of each species is adapted for feeding on a particular type of food. *Geospiza fuliginosa* evolved as an eater of small seeds. *Geospiza magnirostris* evolved on a different island as an eater of large seeds.

(i) Suggest how these different species of finch might have evolved from one species. [3]

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(ii) Describe how biogeography may provide support for the evolution of the different species of finch. [2]

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[Total: 13 marks]

Section B

Answer **only ONE** question.
Write your answer on the foolscap paper provided.

Question 8 [20 mks]

(a) Explain the role of the F plasmids in bacterial conjugation and suggest the advantages of conjugation over transduction. [8]
(b) Differentiate between the viral genome and the eukaryotic genome. [7]
(c) Describe the differences in the life cycle of T4 phage and HIV. [5]

Question 9 [20 mks]

(a) Describe the structural features of a named membranous organelle that is involved in ATP synthesis in photosynthesis. [6]
(b) Compare the two main pathways in which an excited electron can undergo during the light-dependent stage of photosynthesis. [8]
(c) With reference to the term ‘limiting factor’, discuss how two named factors have effects on photosynthesis. [6]

-THE END-